1) Scenario : Event Communication Security

a) Encryption using a shift of 4 positions

The Caesar cipher shifts each letter 4 positions forward in the alphabet.

Original message : MEET AT NOON

Original	M	E	E	T	A	T	N	O	O	N
Encrypted	Q	I	I	X	E	X	R	S	S	R

Encrypted message : QIIX EX RSSR

b) Decryption

The authorized member shifts each encrypted letter 4 positions backward to rebrieve the original message.

Example : Q → M, I → E, X → T

therefore, the original message is MEET AT NOON.

2) Scenario : Interepted suspicious Message

a) Original message

Encrypted message : GSRH RH ZVXIVG

Using the Atbash cipher, each letter is replaced with its opposite letter in the alphabet (A ↔ Z, B ↔ Y, C ↔ X, etc)

Original message : THIS IS A SECRET

b) Justification

* The message contains recognizable word patterns.
* The single-letter word Z becomes A.
* The repeated letters and word lengths match the phrase "THIS IS A SECRET".
* The Atbash cipher uses a fixed reverse-alphabet substitution.

Therefore, the most likely original message is THIS IS A SECRET.

3) Scenario : Secure Messaging System

a) Encryption using the keyword KEY

The Vigenere cipher uses a keyword to determine the shift for each letter.

Plaintext : HELLO

Keyword : KEYKE

Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E
Ciphertext	R	I	J	V	S

Encrypted message : RIJVS

b) Influence of the repeating keyword

* Each keyword letter determines the shift applied t

the corresponding plaintext letter.

* The keyword repeats all message letters are encrypted.
* Different keyword letters produce different shifts.

Therefore, the ciphertext is RIIVS.

4) Scenario : Suspicious Digital Asset Investigation

a) Detecting hidden data in an image

Investigators can use steganalysis, which is the process of detecting hidden information inside digital files.

Steps :

1) Examine the image's file size and metadata.
2) Compare the image with the original, if available.
3) Use steganalysis software to identify unusual pixel patterns.
4) Inspect least significant bits (LSBs) for possible hidden information.
5) Check for unexpected data appended to the image file.

b) Two techniques to strengthen security

1) Encryption : Encrypt the confidential message before embedding it in the image.

2) LSB Steganography : Embed the encrypted data in selected least significant bits of image pixels to make changes less noticeable.

5) Scenario : Legacy Encryption in Corporate Communication.

a) Decryption

"the given ciphertext is :

WECRLTEERDSOEEFEAOCAIVDEN

This resembles a three rail Rail Fence cipher, a transposition cipher that rearranges character across 3 rows.

However, the ciphertext as provided does not clearly produce a meaningful original message under the standard three-rail Rail Fence decryption. The ciphertext or the stated encryption method may contain an error, so the original message cannot be determined reliably without clarification.

b) * It only rearranges letters and does not change the letters themselves.
* Letter frequencies remain uncharged..
* Attackers can test possible arrangements.
* It does not provide modern cryptographic security.

c) Practical enhancement

Replace the legacy transposition cipher with AES (Advanced Encrypted standard) using a secure mode of operation and proper key management. AES provides much stronger confidentiality for modern communication.